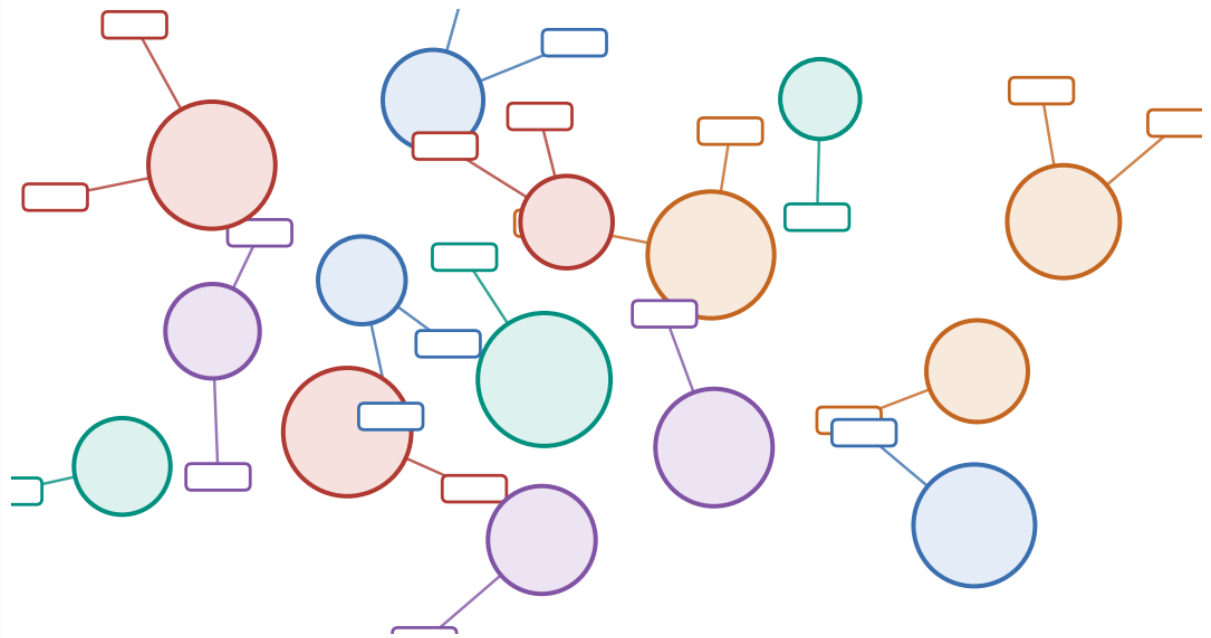
FULL-STACK DATA SCIENCE WITH PYTHON

An illustrated companion to the course notebooks



MODULE 01

Python Foundations for Data Science

The habits that keep your numbers honest

How to read this book

This book walks through every idea in the Module 01 notebooks, but it is written to be *read*, on a train or with a coffee, rather than executed. The notebooks remain the place where you run code, break it and fix it; the book is where you build the mental pictures that make the code obvious.

Each chapter mirrors one notebook and opens with the notebook's file name, so you can always jump between the two. Code in the book is trimmed to the lines that carry the idea, and every figure was generated from code in this repository, using the same colour palette as the notebooks.

Five kinds of boxes appear along the way. They always mean the same thing:

> **NOTE**
> Extra context. Worth knowing, but you can move on without it.

> **TIP**
> A habit or shortcut professionals actually use.

> **WATCH OUT**
> A mistake that produces silently wrong numbers, not an error.

> **FROM THE HISTORY OF DATA SCIENCE**
> A true episode from the history of science and industry that makes the idea stick. Sources are listed at the back.

> **KEY TAKEAWAYS**
> The handful of ideas to carry out of the chapter.

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Sources for the Stories

The history boxes in this book simplify, as stories must, but none of them is invented. These are the sources behind each one, in the order they appear, so that you can read the full versions and check the details for yourself.

Chapter 1

- **The title "data scientist".** T. H. Davenport and D. J. Patil, "Data Scientist: The Sexiest Job of the 21st Century", *Harvard Business Review*, October 2012.
- **John Snow and Broad Street.** J. Snow, *On the Mode of Communication of Cholera*, 2nd ed., London: John Churchill, 1855. For a readable modern account: S. Johnson, *The Ghost Map*, Riverhead, 2006.
- **Reinhart and Rogoff.** C. M. Reinhart and K. S. Rogoff, "Growth in a Time of Debt", *American Economic Review* 100(2), 2010; T. Herndon, M. Ash and R. Pollin, "Does High Public Debt Consistently Stifle Economic Growth? A Critique of Reinhart and Rogoff", *Cambridge Journal of Economics* 38(2), 2014 (working paper 2013).
- **"Student".** Student [W. S. Gosset], "The Probable Error of a Mean", *Biometrika* 6(1), 1908.

Chapter 2

- **Gangnam Style and the 32-bit counter.** YouTube's own announcement, early December 2014, widely reported by the international press at the time.
- **The Vancouver Stock Exchange index.** Reported in *The Wall Street Journal*, November 1983, when the exchange published the recalculated index; since then a standard example in courses on numerical computing.
- **Python's origins.** G. van Rossum, foreword to *Programming Python*, 1st ed., O'Reilly, 1996; and the Python documentation's "General Python FAQ".

Chapter 3

- **Why numbering should start at zero.** E. W. Dijkstra, EWD831, 11 August 1982. Available from the E. W. Dijkstra Archive, University of Texas at Austin.
- **Hashing at IBM.** D. E. Knuth, *The Art of Computer Programming*, Vol. 3: *Sorting and Searching*, 2nd ed., Addison-Wesley, 1998, section 6.4 (history of hashing, crediting H. P. Luhn's January 1953 IBM memorandum).
- **The Federalist Papers.** F. Mosteller and D. L. Wallace, *Inference and Disputed Authorship: The Federalist*, Addison-Wesley, 1964.
- **Timsort.** T. Peters, "listsort.txt", in the CPython source tree, 2002.

Chapter 4

- **Memo functions.** D. Michie, "'Memo' Functions and Machine Learning", *Nature* 218, 1968.
- **The lambda calculus.** A. Church, "An Unsolvable Problem of Elementary Number Theory", *American Journal of Mathematics* 58(2), 1936.
- **MapReduce.** J. Dean and S. Ghemawat, "MapReduce: Simplified Data Processing on Large Clusters", *OSDI*, 2004.

Chapter 5

- **Simula.** O.-J. Dahl and K. Nygaard, "SIMULA: An ALGOL-Based Simulation Language", *Communications of the ACM* 9(9), 1966; ACM A. M. Turing Award citation, 2001.
- **Composition over inheritance.** E. Gamma, R. Helm, R. Johnson and J. Vlissides, *Design Patterns: Elements of Reusable Object-Oriented Software*, Addison-Wesley, 1994.
- **scikit-learn.** F. Pedregosa et al., "Scikit-learn: Machine Learning in Python", *Journal of Machine Learning Research* 12, 2011; L. Buitinck et al., "API design for machine learning software: experiences from the scikit-learn project", ECML PKDD Workshop, 2013.

Chapter 6

- **CSV.** Y. Shafranovich, RFC 4180, "Common Format and MIME Type for Comma-Separated Values (CSV) Files", IETF, October 2005 (Informational).
- **Excel and gene names.** M. Ziemann, Y. Eren and A. El-Osta, "Gene name errors are widespread in the scientific literature", *Genome Biology* 17, 2016; E. A. Bruford et al., "Guidelines for human gene nomenclature", *Nature Genetics* 52, 2020.
- **JSON.** D. Crockford, json.org; ECMA-404, *The JSON Data Interchange Syntax*, 2013.
- **Dremel and Parquet.** S. Melnik et al., "Dremel: Interactive Analysis of Web-Scale Datasets", *Proceedings of the VLDB Endowment* 3(1), 2010; Apache Parquet project history, parquet.apache.org.

Chapter 7

- **The moth.** Logbook of the Harvard Mark II, 9 September 1947, Smithsonian National Museum of American History.
- **Mars Climate Orbiter.** NASA, *Mars Climate Orbiter Mishap Investigation Board Phase I Report*, 10 November 1999.

Chapter 8

- **Public Health England's lost test results.** Public Health England statement of 4 October 2020 and the Health Secretary's statement to Parliament on 5 October 2020; reported widely, including by BBC News, at the time.